THE ASSAM ROYAL GLOBAL UNIVERSITY

ROYAL SCHOOL OF LIBRARY AND INFORMATION SCIENCES (RSLIS)

Department of Library and Information Science, Betkuchi, Guwahati

SYLLABUS & COURSE STRUCTURE

Master of Library and Information Science (MLISC)

2 Year Post-Graduate Programme

With Mapping of

UN Sustainable Development Goals (SDGs)

AY - 2026-27

against Course Outcomes and Detailed Syllabus Units

PROGRAMME STRUCTURE

FIRST SEMESTER

Course Code	Course Title	Level	Credit	L-T-P-C
LIB224C101	Foundation of Library and Information Science (Theory)	400	4	3-1-0-4
LIB224C102	Organization of Knowledge Classification and Cataloging (Theory)	400	4	3-1-0-4
LIB224C113	Organization of Knowledge Classification and Cataloging (Practical)	400	4	0-0-8-4
LIB224C104	Information Source and Services (Theory)	400	4	3-1-0-4
LIB224C105	Foundation of Computer & Information Technology (Theory)	400	4	3-1-0-4
LIB224S106	Management of Library and Information Centre (Swayam)	400	4	

Total Credits for 1st Semester: 24

SECOND SEMESTER

Course Code	Course Title	Level	Credit	L-T-P-C
LIB224C201	Management of Library and Information Centre (Theory)	400	4	3-1-0-4
LIB224C202	Library Automation and Software Package (Theory)	400	4	3-1-0-4
LIB224C213	Library Automation and Software Package (Practical)	400	4	0-0-8-4
LIB224C204	Library System Analysis and Design (Theory)	400	4	3-1-0-4
LIB224C215	Collection Development and Reference Management (Practical)	400	4	0-0-8-4
LIB224S206	Koha Library Management System (Swayam)	400	4	

Total Credits for 2nd Semester: 24 | 1st Year Total: 48

THIRD SEMESTER

Course Code	Course Title	Level	Credit	L-T-P-C
LIB224C301	Information Retrieval (Theory)	400	4	3-1-0-4
LIB224C302	Digital Library Software Packages (Theory)	400	4	3-1-0-4
LIB224C313	Digital Library Software Packages (Practical)	400	4	0-0-8-4
LIB224C404	Research Methodology (Theory)	400	4	3-1-0-4
LIB224C315	Library House Keeping Operations (Practical)	500	4	0-0-8-4

Total Credits for 3rd Semester: 20

FOURTH SEMESTER

Course Code	Course Title	Level	Credit	L-T-P-C
LIB224C401	E-Learning and Content Management System (Practical)	400	4	0-0-8-4
LIB224C402	Digital Resource Management (Theory)	400	4	3-1-0-4
LIB224C413	Digital Resource Management (Practical)	400	4	0-0-8-4
LIB224C304	Database Management System (Theory)	400	4	3-1-0-4
LIB224C405	Media Information Literacy and Copyright (Theory)	500	4	3-1-0-4

Total Credits for 2nd Year: 40 | Programme Total: 88

UN SUSTAINABLE DEVELOPMENT GOALS (SDGs) – LEGEND

SDG	Goal	Description
SDG 4	**Quality Education**	Ensure inclusive and equitable quality education and promote lifelong learning
SDG 8	**Decent Work & Economic Growth**	Promote sustained, inclusive economic growth and productive employment
SDG 9	**Industry, Innovation & Infrastructure**	Build resilient infrastructure, promote inclusive industrialization and foster innovation
SDG 10	**Reduced Inequalities**	Reduce inequality within and among countries
SDG 11	**Sustainable Cities & Communities**	Make cities inclusive, safe, resilient and sustainable
SDG 12	**Responsible Consumption & Production**	Ensure sustainable consumption and production patterns
SDG 16	**Peace, Justice & Strong Institutions**	Promote peaceful societies, provide access to justice for all
SDG 17	**Partnerships for the Goals**	Strengthen implementation and revitalize global partnership for sustainable development

SEMESTER I

First Semester Courses

Master of Library and Information Science (MLISC)

LIB224C101 L-T-P-C: 3-1-0-4 | Credit: 4 Evaluation: Theory (T)	FOUNDATION OF LIBRARY AND INFORMATION SCIENCE

COURSE OBJECTIVE

The objective of the course is to introduce the students with basic philosophy of Library and Information Science.

COURSE OUTCOMES

CO. No.	Course Outcome	SDG Goals	Bloom's Taxonomy Level
CO1	Know the contribution of Dr. S.R. Ranganathan in the field of LIS and get acquainted with the Five Laws of Library Science.	SDG 4: Quality Education	BT Level I
CO2	Demonstrate the role of libraries and its importance in human civilization and understand the history, development and current role of libraries in society.	SDG 4: Quality Education SDG 11: Sustainable Cities	BT Level II
CO3	Compare between Public, Academic and Special Libraries and their functions and identify different types of libraries and their unique functions.	SDG 4: Quality Education SDG 10: Reduced Inequalities	BT Level II
CO4	Make acquainted with the roles and responsibilities of professional associations in LIS profession.	SDG 17: Partnerships for the Goals	BT Level III

DETAILED SYLLABUS

Module	Topics and Course Contents	SDG Goals	Periods
Unit 1	Library - Concepts, Definition and Functions; Library as a social institution; Library's role in Information exchange, Recreation and Community Information; Information Industry - Generators, Providers and Intermediaries; Dr. S.R. Ranganathan: An Introduction; Contribution of Dr. S.R. Ranganathan in Library Science; Five Laws of Library and Information Science and their implications	SDG 4: Quality Education SDG 16: Peace, Justice & Strong Institutions	15
Unit 2	Types of Library: Objectives, Features and Functions; Public Library; Academic Library; Special Library; Historical Development of Library in the World; Library Movement in India with special reference to North East India; Librarianship: Professional Ethics; Library Legislation in India with Special Reference to North East India; Library Extension Services; Community Information Services; Intellectual Property Rights (IPR) - Concept and Types	SDG 4: Quality Education SDG 10: Reduced Inequalities SDG 16: Peace, Justice & Strong Institutions	15
Unit 3	Library Association: Meaning, Objectives and Functions, Types; ALA; ILA; IASLIC; IATLIS; LA (UK); IFLA; CILIP	SDG 17: Partnerships for the Goals	15
Unit 4	UNESCO - Objectives, activities and services; UNESCO and Public Library Manifesto; RRRLF - Objectives, Functions; Public Libraries of India: Asiatic Public Library, Khuda Bakash Oriental Library, Baroda Public Library System, Connemara Public Library, Delhi Public Library	SDG 4: Quality Education SDG 11: Sustainable Cities SDG 17: Partnerships for the Goals	15
	TOTAL PERIODS		**60**

TEXTBOOKS

R.S. Prajapati (2013). Foundations of Library and Information Science, Discovery Publishing House, New Delhi.
Bauer, Patricia. (2020). Foundations of library and information science. New York, Neal-Schuman.

REFERENCE BOOKS

Agarwal, U. K. (1999). Twentieth century: Library legislation in India, Udaipur, Shiva Publishers.
Carlson, Christ, & Brosnahan, Ellen. (2008). Guiding students into information literacy. USA, Scarecrow Press.
Crawford, Walt. (1998). Being analog: Creating tomorrow's libraries, Chicago, ALA.

Ismail, Abdullah. (2009). Global library and information science. New York, Walter de Gruyter.
Leckie, Gloria J, et al. (2010). Critical theory for library and information science, California, Libraries Unlimited.
Maskus, Keith E. (2000). Intellectual property rights in the global economy, Washington DC.
Panella, Deborah, & Mount, Ellis (2012). Basics of law librarianship, New York, Routledge.
Ramage, Magnus & Chapman, David. (2011). Perspectives on information, New York, Routledge.

LIB224C102 L-T-P-C: 3-1-0-4 | Credit: 4 Evaluation: Theory (T)	ORGANIZATION OF KNOWLEDGE: CLASSIFICATION AND CATALOGING

COURSE OBJECTIVE

The objective of the course is to give hands-on knowledge and skills in Library classification and cataloguing.

COURSE OUTCOMES

CO. No.	Course Outcome	SDG Goals	Bloom's Taxonomy Level
CO1	Define the basics of library classification and practical implication of the major classification schemes.	SDG 4: Quality Education	BT Level I
CO2	Demonstrate a comprehensive understanding of cataloguing principles and standards, including bibliographic description, subject analysis, classification and authority control.	SDG 4: Quality Education SDG 9: Industry & Innovation	BT Level II
CO3	Identify the fundamentals of classification schemes in organizing knowledge resources.	SDG 4: Quality Education	BT Level III
CO4	Classify the fundamental skills of Library cataloguing system.	SDG 4: Quality Education SDG 9: Industry & Innovation	BT Level IV

DETAILED SYLLABUS

Module	Topics and Course Contents	SDG Goals	Periods
Unit 1	Library Classification: Origin, Meaning, Need and Purpose; History of Classification; Universe of Knowledge: Structures and attributes; Modes of formation of subjects	SDG 4: Quality Education	15
Unit 2	Normative Principles of Classification: General Theory; APUPA; Types and Features of Classification Schemes; DDC, UDC, LCC, CC; Universe of subjects as mapped in major classification schemes	SDG 4: Quality Education SDG 9: Industry & Innovation	15
Unit 3	Library Catalogues: Fundamental Concepts; Historical Developments; Definition, Objectives, Purposes and Functions; Trends; Centralized and Cooperative Cataloguing; ISBD, MARC, CCF, ISBN, ISSN	SDG 4: Quality Education SDG 9: Industry & Innovation	15
Unit 4	Idea plane: canons, principles and postulates; fundamental categories; Verbal plane; Notation; Subject heading: SLSH, LCSH; Chain indexing; AACR II and Classified Cataloguing Code	SDG 4: Quality Education SDG 9: Industry & Innovation	15
	TOTAL PERIODS		**60**

TEXTBOOKS

Krishan Kumar (2023). Theory of cataloguing. New Delhi: Vikas Publication.
Krishan Kumar (2023). Theory of classification. New Delhi: Vikas Publication.

REFERENCE BOOKS

Carter, R.C. ed. (2001). Managing cataloguing and the organization of information. New York: Haworth Press.
Dhiman, A.K. & Yashoda Rani. (2005). Learn library classification. New Delhi: Ess Ess.
Kao, Mary L. (2001). Cataloguing and classification for library technicians. New York: Haworth Press.

LIB224C113 L-T-P-C: 0-0-8-4 | Credit: 4 Evaluation: Practical (P)	ORGANIZATION OF KNOWLEDGE: CLASSIFICATION AND CATALOGING (PRACTICAL)

COURSE OBJECTIVE

The objective of the course is to give practical knowledge and skills in Library classification using DDC, UDC and Colon classification.

COURSE OUTCOMES

CO. No.	Course Outcome	SDG Goals	Bloom's Taxonomy Level
CO1	Demonstrate a comprehensive understanding of the historical development, principles, and structure of DDC system.	SDG 4: Quality Education	BT Level II
CO2	Demonstrate an in-depth understanding of the historical development, principles, and theoretical foundations of Colon Classification (CC).	SDG 4: Quality Education	BT Level II
CO3	Develop skills in subject analysis and proficiency in using standard schemes of classification.	SDG 4: Quality Education SDG 9: Industry & Innovation	BT Level III
CO4	Apply DDC principles effectively to classify a wide array of knowledge domains and information resources.	SDG 4: Quality Education	BT Level IV

DETAILED SYLLABUS

Module	Topics and Course Contents	SDG Goals	Periods
Unit 1	Classification of Documents representing basic, Compound and Complex Subject according to CC (Colon Classification)	SDG 4: Quality Education	15
Unit 2	Classification of Documents requiring common subdivisions and other auxiliaries; Classification of documents according to DDC (23rd edition)	SDG 4: Quality Education	15
Unit 3	Cataloguing of works of personal authors: Shared Responsibility, Mixed Responsibility, Anonymous works, corporate works, Serial publications, Non-Book Materials, Internet and Multimedia Resources; According to AACR-2	SDG 4: Quality Education SDG 9: Industry & Innovation	15
Unit 4	Assigning subject heading Using SLSH and LCSH; Anonymous works; Works of corporate authorship and Analytical	SDG 4: Quality Education	15
	TOTAL PERIODS		**60**

TEXTBOOKS

Melvil Dewey et al. (2023). Dewey Decimal Classification and Relative Index, 23rd edition. OCLC.
S. R. Ranganathan (2023). Colon Classification Sixth Edition. Sikhar Publishing House.

REFERENCE BOOKS

Dhyani, Pushpa. (2006). Classifying with Dewey decimal classification. New Delhi: Ess Ess.
Kaula, P.N. (1985). A treatise on colon classification. New Delhi: Sterling Publishers.
Singh, S.N. & Prasad, H.N. (1985). Cataloguing manual AACR-II. Delhi: B.R. Publishing.

LIB224C104 L-T-P-C: 3-1-0-4 | Credit: 4 Evaluation: Theory (T)	INFORMATION SOURCES AND SERVICES

COURSE OBJECTIVE

The objective of the course is to acquire knowledge on reference and information sources and services in context of libraries.

COURSE OUTCOMES

CO. No.	Course Outcome	SDG Goals	Bloom's Taxonomy Level
CO1	Define planning, managing, and evaluating information services in different settings, including libraries, information centers, and online platforms.	SDG 4: Quality Education SDG 9: Industry & Innovation	BT Level I
CO2	Demonstrate the principles and practices of reference services, including techniques for reference interviews, tools, and methods.	SDG 4: Quality Education	BT Level II
CO3	Develop abilities to teach information literacy skills to users, including effective search strategies, source evaluation and citation practices.	SDG 4: Quality Education SDG 16: Peace, Justice & Strong Institutions	BT Level III
CO4	Classify the significance of information literacy, educating users on information seeking, evaluation, and ethical use of information resources.	SDG 4: Quality Education SDG 10: Reduced Inequalities	BT Level IV

DETAILED SYLLABUS

Module	Topics and Course Contents	SDG Goals	Periods
Unit 1	Sources of Information: Primary (Periodicals, Conferences, Patents, Standards, Theses, Trade Literature); Secondary (Dictionaries, Encyclopaedias, Biographical, Geographical, Bibliographies, Indexing and Abstracting); Tertiary (Directories, Yearbooks, Almanacs, Union Catalogues); Criteria for evaluation	SDG 4: Quality Education	15
Unit 2	Reference Service – Concepts, Scope, Modes, Enquiry Techniques; Information Searching from Print and Electronic Sources; Referral Services; Document Delivery; Translation Services; Role of Reference Staff	SDG 4: Quality Education SDG 9: Industry & Innovation	15
Unit 3	Information Needs and Information Seeking Behaviour; Models by Wilson, Dervin, Kuhlthau and Ellis; Information Sources and Services in Humanities, Social Sciences, Sciences, Business and Law	SDG 4: Quality Education SDG 10: Reduced Inequalities	15
Unit 4	User Studies – History, Concepts, Goals, Methodology; User Education; Documentation Services: Abstracting, Indexing, CAS, SDI, Reprographic Service, Translation, Document Delivery, Bulletin Board, Referral and Digital Reference Service	SDG 4: Quality Education SDG 9: Industry & Innovation	15
	TOTAL PERIODS		**60**

TEXTBOOKS

Connor, E. (2006). An introduction to reference services in academic libraries. New York: Haworth.
Katz, B. (2003). Digital reference services. Binghamton, NY: Haworth Information Press.

REFERENCE BOOKS

Case, D.O. (2002). Looking for Information. California: Academic Press.
Chowdhury, G. G. (2011). Information users and usability in the digital age. New York: Neal-Schuman.
Kern, M. K. (2009). Virtual reference best practices. Chicago: ALA.

LIB224C105 L-T-P-C: 3-1-0-4 | Credit: 4 Evaluation: Theory (T)	FOUNDATION OF COMPUTER & INFORMATION TECHNOLOGY

COURSE OBJECTIVE

The objective of the course is to give the knowledge on library automation and Networking.

COURSE OUTCOMES

CO. No.	Course Outcome	SDG Goals	Bloom's Taxonomy Level
CO1	Define the basic components of a computer system, including hardware (CPU, memory, storage), software (OS, applications), and their functionalities.	SDG 4: Quality Education SDG 9: Industry & Innovation	BT Level I
CO2	Compare operating system functionalities such as process management, memory management, file systems and user interfaces.	SDG 9: Industry & Innovation	BT Level II
CO3	Identify the fundamental concepts of computer networks, including types, topologies, protocols and the Internet.	SDG 9: Industry & Innovation	BT Level III
CO4	Identify a comprehensive understanding of various transmission media, including guided (wired) and unguided (wireless) channels.	SDG 9: Industry & Innovation	BT Level III

DETAILED SYLLABUS

Module	Topics and Course Contents	SDG Goals	Periods
Unit 1	Computer Applications: introduction, development, generations; Information Technology: introduction and scope; Computer Components: Hardware and Software, I/O Devices; Storage devices; Number system: Binary, ASCII, Unicode; Data types	SDG 4: Quality Education SDG 9: Industry & Innovation	15
Unit 2	Operating System: concept, types and functions (DOS, LINUX, Windows); File formats; Office Packages: Word Processor, Spreadsheet, Presentation Tools, MS-Access	SDG 9: Industry & Innovation	15
Unit 3	Telecommunication: Transmission Channels, Mode and Media; Multiplexing, Modulation, Standards (ISDN, PSDN); Wireless Communication: Wi-Fi, Li-Fi, Satellite, Mobile Communication	SDG 9: Industry & Innovation	15
Unit 4	Fundamentals of Internet: History, OSI Model; Network Types, Topology; WWW: Introduction, History, Recent Developments; Search Engine, Meta Search Engine; Semantic Web	SDG 4: Quality Education SDG 9: Industry & Innovation	15
	TOTAL PERIODS		**60**

TEXTBOOKS

Sinha, P.K (2004). Computer Fundamentals. 6th Ed. BPB Publications: New Delhi.
Ram, B (2007). Computer Fundamentals: Architecture and Organization. 4th Ed. New Age International.

REFERENCE BOOKS

Rajaraman, V. Fundamentals of Computers. 5th Ed. Prentice Hall India, 2010.
Bansal, S.K. Information Technology and Globalization, New Delhi: A.P.H. Publishing, 2005.

SEMESTER II

Second Semester Courses

Master of Library and Information Science (MLISC)

LIB224C201 L-T-P-C: 3-1-0-4 | Credit: 4 Evaluation: Theory (T)	MANAGEMENT OF LIBRARY AND INFORMATION CENTRE

COURSE OBJECTIVE

The objective of the course is to acquire knowledge on Management purposes in Library and Information Centre.

COURSE OUTCOMES

CO. No.	Course Outcome	SDG Goals	Bloom's Taxonomy Level
CO1	Define knowledge of the principles, theories, and practices related to library and information center management.	SDG 4: Quality Education SDG 16: Peace, Justice & Strong Institutions	BT Level I
CO2	Explain a comprehensive understanding of Total Quality Management, including its history, evolution and key components.	SDG 9: Industry & Innovation SDG 12: Responsible Consumption	BT Level II
CO3	Explain the key principles, theories, and functions of Human Resource Management within organizational contexts.	SDG 8: Decent Work & Economic Growth	BT Level III
CO4	Develop effective communication skills to articulate SWOT analysis findings.	SDG 8: Decent Work & Economic Growth SDG 9: Industry & Innovation	BT Level III

DETAILED SYLLABUS

Module	Topics and Course Contents	SDG Goals	Periods
Unit 1	Management Concept, Functions and Principles; Schools of Management Thought: Classical, Scientific, Behavioural, Decision Theory, Contingency Approach, Systems Approach	SDG 4: Quality Education SDG 16: Peace, Justice & Strong Institutions	15
Unit 2	Planning: Concept, Need and Levels; MBO; Decision Making; Total Quality Management (TQM); Change Management	SDG 9: Industry & Innovation SDG 12: Responsible Consumption	15
Unit 3	Human Resource Management: Manpower Planning; Job Analysis, Description and Evaluation; Recruitment; Performance Appraisal; Leadership; Communication Process; Motivation; Organizational Manual; Annual Report	SDG 8: Decent Work & Economic Growth	15
Unit 4	Budgeting – Concept, Principles and Types; Resource Mobilisation; Marketing of Information Products and Services; SWOT Analysis; Project Management	SDG 8: Decent Work & Economic Growth SDG 9: Industry & Innovation	15
	TOTAL PERIODS		**60**

TEXTBOOKS

Chabhra, T. N. et al. (2000). Management and organisation. New Delhi: Vanity Book International.

P. Balasubramanian (2021). Management of Libraries and Information Centers. Ess Ess Publication.

REFERENCE BOOKS

Bryson, Jo. (2011). Managing information services: A sustainable approach. England: Ashgate.

Daft, Richard L. & Marcic, Dorothy. (2012). Understanding management. 8th edition.

LIB224C202 L-T-P-C: 3-1-0-4 | Credit: 4 Evaluation: Theory (T)	LIBRARY AUTOMATION SOFTWARE PACKAGE

COURSE OBJECTIVE

This is to train the students on Open-Source Library Management Software and Software.

COURSE OUTCOMES

CO. No.	Course Outcome	SDG Goals	Bloom's Taxonomy Level
CO1	Define theoretical knowledge about the concept, evolution, and types of library automation systems, including ILS and LMS.	SDG 4: Quality Education SDG 9: Industry & Innovation	BT Level I
CO2	Explain theoretical frameworks and standards such as MARC, Z39.50 and other industry standards used in library automation.	SDG 9: Industry & Innovation	BT Level II
CO3	Explain theoretical discussions on emerging trends, innovations and future developments in library automation software.	SDG 9: Industry & Innovation	BT Level II
CO4	Apply GNU licensing to software projects, ensuring compliance with open-source principles.	SDG 9: Industry & Innovation SDG 17: Partnerships for the Goals	BT Level III

DETAILED SYLLABUS

Module	Topics and Course Contents	SDG Goals	Periods
Unit 1	Open-Source Software: Overview, Philosophy; Commercial Software, Free Software, Freeware; NISO and DLF Standards; Metadata: Dublin Core, MARC, RDA; OAI-PMH; Licensing: GNU, Open-Source Licenses	SDG 9: Industry & Innovation SDG 17: Partnerships for the Goals	15
Unit 2	Library standards – scope, objectives, types; Library automation standards; Open standards; Library automation software – functional requirements, RFPs; ILSs in India; Open-source software in library automation	SDG 9: Industry & Innovation	15
Unit 3	Library system and subsystems; Procedural model of library automation; Software-level requirements; Components: Acquisition, Cataloguing, Circulation and Serials control subsystems	SDG 9: Industry & Innovation	15
Unit 4	Trends and future: Web-scale discovery, linked open data, Cloud-based library automation, Library mashup	SDG 9: Industry & Innovation	15
	TOTAL PERIODS		**60**

REFERENCE BOOKS

Breeding, Marshall. (2010). LibLime Acquisition by PTFS Marks a New Era for Koha. Library Journal.

Eyler, Pat. (2003). Koha: a gift to libraries from New Zealand. Linux Journal.

LIB224C213 L-T-P-C: 0-0-8-4 | Credit: 4 Evaluation: Practical (P)	LIBRARY AUTOMATION SOFTWARE PACKAGE (PRACTICAL)

COURSE OBJECTIVE

This is to train the students on Open-Source Library Management Software and Software.

COURSE OUTCOMES

CO. No.	Course Outcome	SDG Goals	Bloom's Taxonomy Level
CO1	Define practical knowledge of different library automation software packages, their features, functionalities and how they support library operations.	SDG 4: Quality Education SDG 9: Industry & Innovation	BT Level I
CO2	Develop skills in implementing library automation systems, including installation, configuration and customization.	SDG 9: Industry & Innovation	BT Level II
CO3	Develop managing library databases within the automation software, including cataloging, indexing and maintaining accurate records.	SDG 9: Industry & Innovation	BT Level III
CO4	Develop best practices for data security, backup procedures and disaster recovery plans.	SDG 9: Industry & Innovation SDG 16: Peace, Justice & Strong Institutions	BT Level IV

DETAILED SYLLABUS

Module	Topics and Course Contents	SDG Goals	Periods
Unit 1	Open-Source OS (e.g., Linux) Hosting: Client, Server; Library Management Software: Proprietary and OSS; Introduction to SOUL; LMS Koha	SDG 9: Industry & Innovation	10
Unit 2	Practical's on Koha: Introduction, Features, Architecture, Installation, Customization; Modules: Acquisition, Cataloguing, Serials control, Circulation and Patron Management; Reports, Export/Import, Tools	SDG 4: Quality Education SDG 9: Industry & Innovation	20
Unit 3	Practical's on SOUL: Introduction, Features, Architecture, Installation, Customization; Modules: Acquisition, Cataloguing, Serials control, Circulation and Patron Management; Reports, Export/Import	SDG 4: Quality Education SDG 9: Industry & Innovation	20
Unit 4	ILS Administration and User Interface: OPAC, OPAC 2.0, OPAC Mashup; Usage statistics through Web analytics; Enhanced features; Privilege control; Backup and Restoration	SDG 9: Industry & Innovation	10
	TOTAL PERIODS		**60**

REFERENCE BOOKS

Breeding, Marshall. (2002). An Update on Open Source ILS. Information Today.
Kumar V (2008). Selection and management of open source software in libraries.

LIB224C204 L-T-P-C: 3-1-0-4 | Credit: 4 Evaluation: Theory (T)	LIBRARY SYSTEM ANALYSIS AND DESIGN

COURSE OBJECTIVE

The paper will enable students to develop and manage information systems in libraries using: Information systems concepts, Requisite Skills, System Methodologies, Development Tools and Perspectives on system development.

COURSE OUTCOMES

CO. No.	Course Outcome	SDG Goals	Bloom's Taxonomy Level
CO1	Define the library system as well as its various sub-components or sections.	SDG 4: Quality Education	BT Level I
CO2	Compare the knowledge of interacting entities, including computer systems analysis, different tools and techniques.	SDG 9: Industry & Innovation	BT Level II
CO3	Identify the effective packaging and repackaging of information for effective dissemination.	SDG 9: Industry & Innovation SDG 4: Quality Education	BT Level III
CO4	Utilize the skill about information need, information seeking behaviour in both human and technological contexts; user studies and education.	SDG 4: Quality Education SDG 9: Industry & Innovation	BT Level IV

DETAILED SYLLABUS

Module	Topics and Course Contents	SDG Goals	Periods
Unit 1	Information Systems: Types (TPS, MIS, DSS); System Development Life Cycle; Development Process: Waterfall, Prototyping, CASE, JAD, RAD, Agile and eXtreme Programming	SDG 9: Industry & Innovation	15
Unit 2	Systems Acquisition; Outsourcing; Off-the-Shelf Software; RFP and AMC; Feasibility Assessment: Economic, Technical and Other; Baseline Project Plan	SDG 9: Industry & Innovation	15
Unit 3	System Implementation: Coding, Testing, Installation, Documentation, Training; System Maintenance: Types, Cost; System Analyst: Role, Responsibilities, Skills; Security and Ethics	SDG 9: Industry & Innovation SDG 16: Peace, Justice & Strong Institutions	15
Unit 4	Informetrics, Bibliometrics, Scientometrics, Altmetrics and Webometrics; Bibliometric Laws: Lotka's, Bradford's and Zipf's Law; Citation Analysis, Co-citation Coupling and Bibliographic Coupling	SDG 4: Quality Education SDG 9: Industry & Innovation	15
	TOTAL PERIODS		**60**

TEXTBOOKS

Avison, D. E. and Guy Fitzgerald (2006). Information Systems Development. 4th Ed. McGraw-Hill.
Hoffer, Jeffrey A., et al. (2013). Modern Systems Analysis and Design. 7th Ed. Pearson.

REFERENCE BOOKS

Davis, William S. and David C. Yen (2010). The Information System Consultant's Handbook. CRC Press.
Dennis, Alan, et al. (2008). Systems Analysis and Design. 4th Ed. John Wiley & Sons.

LIB224C215 L-T-P-C: 0-0-8-4 | Credit: 4 Evaluation: Practical (P)	COLLECTION DEVELOPMENT AND REFERENCE MANAGEMENT

COURSE OBJECTIVE

The subject provides skills on collection development and hands-on practice of reference management.

COURSE OUTCOMES

CO. No.	Course Outcome	SDG Goals	Bloom's Taxonomy Level
CO1	Define the principles and strategies involved in building, evaluating and managing collections of various information resources.	SDG 4: Quality Education SDG 12: Responsible Consumption	BT Level I
CO2	Compare with various reference tools, databases, bibliographic sources and digital repositories.	SDG 4: Quality Education SDG 9: Industry & Innovation	BT Level II
CO3	Utilize technology tools and software for managing references, bibliographies and citations.	SDG 9: Industry & Innovation	BT Level III
CO4	Compare the various tools and features offered by Reference Management System.	SDG 9: Industry & Innovation	BT Level IV

DETAILED SYLLABUS

Module	Topics and Course Contents	SDG Goals	Periods
Unit 1	Collection Development – Concept, Goals, Methods; Principles by Ranganathan, Drury, Dewey, LC and ALA; Policies; Planning: Committees, Staffing, Budgeting, Implementation and Evaluation	SDG 4: Quality Education SDG 12: Responsible Consumption	15
Unit 2	Selection Tools: Bibliographies, Publishers' Catalogues, Book Reviews; Stock Verification and Rectification; Preservation of Collection (Print and Non-Print)	SDG 4: Quality Education SDG 11: Sustainable Cities	15
Unit 3	Reference Management System – Overview, Concept; Hands-on Practice in Mendeley: Tools, Features, Installation, Integration, Customization	SDG 4: Quality Education SDG 9: Industry & Innovation	15
Unit 4	Citation: concept, need and purpose; Online citation tools; Authority files; Hands-on Practice in Zotero: Tools, Features, Installation, Integration, Customization	SDG 4: Quality Education SDG 9: Industry & Innovation	15
	TOTAL PERIODS		**60**

TEXTBOOKS

Vicki L. Gregory (2019). Collection Development and Management for 21st Century Library Collections. ALA Neal-Schuman.

Abdul Mannan Khan (2023). Collection Development Its Organization And Services. Ess Ess Publications.

REFERENCE BOOKS

Kovacs, B. (1990). The Decision-Making Process for Library Collections. New York: Greenwood Press.

SEMESTER III

Third Semester Courses

Master of Library and Information Science (MLISC)

LIB224C301 L-T-P-C: 3-1-0-4 | Credit: 4 Evaluation: Theory (T)	INFORMATION RETRIEVAL

COURSE OBJECTIVE

The objective of the course is to acquire knowledge on retrieval of document and information storage.

COURSE OUTCOMES

CO. No.	Course Outcome	SDG Goals	Bloom's Taxonomy Level
CO1	Define knowledge on Information Retrieval with search strategies and Academic Databases.	SDG 4: Quality Education	BT Level I
CO2	Explain about information retrieval in the context of LIS with its basic models, techniques and strategies.	SDG 4: Quality Education SDG 9: Industry & Innovation	BT Level II
CO3	Apply different tools, Vocabulary Control and its practical usage in information handling and dissemination.	SDG 9: Industry & Innovation	BT Level III
CO4	Classify different Indexing techniques being employed by libraries of the contemporary era.	SDG 9: Industry & Innovation	BT Level IV

DETAILED SYLLABUS

Module	Topics and Course Contents	SDG Goals	Periods
Unit 1	IR: Concept, Nature; Content Analysis; Vocabulary Control; Subject Indexing: LCSH, Sear's List, Thesaurus, Thesaurofacet, Classaurus, Automatic Indexing, COMPASS	SDG 4: Quality Education SDG 9: Industry & Innovation	15
Unit 2	Indexing languages; Index construction; Theories of Indexing (Rationalist, Historicist, Hermeneutical, Pragmatic, Critical); Pre-coordinate Indexing (Chain, PRECIS, POPSI, SLIC); Post-coordinate (Uniterm, Title derived)	SDG 4: Quality Education SDG 9: Industry & Innovation	15
Unit 3	Search Types: Boolean, Proximity, Fuzzy, Iterative; Dictionaries, Querying, Wildcard, Full Text Search, Spelling/Phonetic Correction; Search Engines; Z39.50; Metadata in IR	SDG 9: Industry & Innovation	15
Unit 4	Design and Evaluation of IRS; IR Models: Probabilistic, Language Models, XML Retrieval; Text Classification, Naive Bayes, Clustering, Web Crawling, Link Analysis; Emerging Trends: AI, Expert Systems, OCR, Text Summarization	SDG 9: Industry & Innovation	15
	TOTAL PERIODS		**60**

TEXTBOOKS

Baeza-Yates, R. A. & Ribeiro-Neto, B. (2010). Modern Information Retrieval. 2nd Ed. Addison-Wesley.
Manning, C. D. et al. (2008). Introduction to Information Retrieval. Cambridge University Press.

REFERENCE BOOKS

Grossman, David A. & Ophir Frieder. (2004). Information Retrieval: Algorithms and Heuristic. 2nd Ed. Springer.

LIB224C302 L-T-P-C: 3-1-0-4 | Credit: 4 Evaluation: Theory (T)	DIGITAL LIBRARY SOFTWARE PACKAGE

COURSE OBJECTIVE

This paper will articulate an understanding of the conceptual and pragmatic basis of digital libraries in the framework of traditional library activities and services.

COURSE OUTCOMES

CO. No.	Course Outcome	SDG Goals	Bloom's Taxonomy Level
CO1	Define comprehensive understanding of the principles, theories, and concepts underlying digital libraries, including architecture and functionalities.	SDG 4: Quality Education SDG 9: Industry & Innovation	BT Level I
CO2	Explain to curate and manage digital collections, including digitization, metadata creation and preservation strategies.	SDG 4: Quality Education SDG 9: Industry & Innovation	BT Level II
CO3	Apply principles of digital preservation, including long-term storage, migration strategies and sustainability of digital collections.	SDG 9: Industry & Innovation SDG 11: Sustainable Cities	BT Level III
CO4	Simplify understanding of workflow concepts and their significance in streamlining business processes.	SDG 9: Industry & Innovation	BT Level IV

DETAILED SYLLABUS

Module	Topics and Course Contents	SDG Goals	Periods
Unit 1	Digital Libraries and Institutional Repositories: Definitions, Concepts, Benefits, Limitations; Planning and Implementation; Hybrid Libraries, Library Portals; IR: Planning, Content Submission	SDG 4: Quality Education SDG 9: Industry & Innovation	15
Unit 2	Digital Formats: Textual and Non-textual; Character Encoding; Mark-ups: Procedural, Presentational, Descriptive; Electronic Image; Image Compression; PDF: features, applications; Document Conversion	SDG 9: Industry & Innovation	15
Unit 3	Infrastructural Requirements; Workflow: Scanning, OCR, Editing; Digitization for Preservation; Metadata: Types, Standards, Harvesting; DL Website: Visibility, Accessibility, Searchability	SDG 9: Industry & Innovation SDG 11: Sustainable Cities	15
Unit 4	Digitization and collection development (hardware, software, process, policies); Collection development best practices; Federated search; Web-scale resource discovery; Semantic web and digital libraries	SDG 4: Quality Education SDG 9: Industry & Innovation	15
	TOTAL PERIODS		**60**

TEXTBOOKS

Arms, William Y. (2001). Digital Libraries. 2nd Ed. MIT Press.
Lesk, Michael. (2005). Understanding Digital Libraries. 2nd Ed. Elsevier.

REFERENCE BOOKS

Borgman, Christine L. (2000). From Gutenberg to the Global Information Infrastructure. MIT Press.
Kresh, Diane (2007). The Whole Digital Library Handbook. ALA.

LIB224C313 L-T-P-C: 0-0-8-4 | Credit: 4 Evaluation: Practical (P)	DIGITAL LIBRARY SOFTWARE PACKAGE (PRACTICAL)

COURSE OBJECTIVE

To provide basic concepts related to digital library software with hands-on practice.

COURSE OUTCOMES

CO. No.	Course Outcome	SDG Goals	Bloom's Taxonomy Level
CO1	Define the concept, significance, and evolution of digital libraries in modern information management.	SDG 4: Quality Education	BT Level I
CO2	Compare the user interface of digital library software, demonstrating competence in accessing features.	SDG 9: Industry & Innovation	BT Level II
CO3	Build a digital library project using the software package, demonstrating proficiency in real-world application.	SDG 9: Industry & Innovation SDG 4: Quality Education	BT Level III
CO4	Discover user-friendly interfaces and provide value-added services to enhance user experience.	SDG 4: Quality Education SDG 9: Industry & Innovation	BT Level IV

DETAILED SYLLABUS

Module	Topics and Course Contents	SDG Goals	Periods
Unit 1	Overview of DSpace: History, Concept, Advantages, Metadata; Overview of Greenstone/E-Print: History, Implementation, Advantage, Metadata	SDG 4: Quality Education SDG 9: Industry & Innovation	10
Unit 2	Metadata management; Collection building and delegation; Multilingual data management; Access and Usage management; Import/Export of Metadata	SDG 9: Industry & Innovation	10
Unit 3	Hands-on Practice DSpace: Installation, User Interface, Collection Building, Uploading, Previewing	SDG 9: Industry & Innovation	20
Unit 4	Hands-on Practice Greenstone/E-Print: Installation, User Interface, Collection Building, Uploading, Previewing	SDG 9: Industry & Innovation	20
	TOTAL PERIODS		**60**

TEXTBOOKS

Arms, William Y. (2001). Digital Libraries. 2nd Ed. MIT Press.

Lesk, Michael. (2005). Understanding Digital Libraries. 2nd Ed. Elsevier.

REFERENCE BOOKS

Witten, Ian H. et al. (2009). How to Build a Digital Library. 2nd Ed. Morgan Kaufmann.

LIB224C304 L-T-P-C: 3-1-0-4 | Credit: 4 Evaluation: Theory (T)	RESEARCH METHODOLOGY

COURSE OBJECTIVE

The objective of the course is to acquire knowledge on research and research methodology.

COURSE OUTCOMES

CO. No.	Course Outcome	SDG Goals	Bloom's Taxonomy Level
CO1	Define research and its significance in various disciplines.	SDG 4: Quality Education	BT Level I
CO2	Explain the fundamental concepts of research.	SDG 4: Quality Education	BT Level II
CO3	Identify the fundamental principles underlying research, including its purpose, process, and various approaches.	SDG 4: Quality Education SDG 16: Peace, Justice & Strong Institutions	BT Level III
CO4	Compare different citation styles commonly used in academic writing (e.g., APA, MLA, Chicago, Harvard).	SDG 4: Quality Education	BT Level IV

DETAILED SYLLABUS

Module	Topics and Course Contents	SDG Goals	Periods
Unit 1	Research: meaning, need, significance; Types: descriptive, analytical, applied, fundamental, quantitative, qualitative, conceptual, empirical; Problem identification; Variables; Hypothesis; Research Proposal	SDG 4: Quality Education	15
Unit 2	Methods: Historical, Scientific, Experimental, Descriptive, Survey, Case Study; Data collection tools: Questionnaire, Schedule, Interview, Observation	SDG 4: Quality Education	15
Unit 3	Data Analysis: Descriptive Statistics, Measures of Central Tendency, Dispersion, Variance, Standard Deviation; Graphical presentation; Parametric and Non-Parametric tests; Statistical Packages	SDG 4: Quality Education SDG 9: Industry & Innovation	15
Unit 4	Citation and Referencing: APA, Chicago, IEEE, MLA; Technical Writing; Writing Research plans/proposals; Research reports: structure, style, guidelines	SDG 4: Quality Education	15
	TOTAL PERIODS		**60**

TEXTBOOKS

Krishan Kumar (1992). Research Method in Library and Information Science. Delhi: Har-Anand.
Kothari, C R (2008). Research Methodology: Methods and Techniques. New Delhi: New Age International.

REFERENCE BOOKS

Borgman, Christine L. (1990). Scholarly Communication and Bibliometrics. Newbury Park: Sage.
Trochim, William (2002). Research Methods Knowledge Base. 2nd ed. Atomic Dog Publishing.
Vaughan, Liwen. (2001). Statistical methods for Information professionals. NJ: Information Today.

LIB224C315 L-T-P-C: 0-0-8-4 | Credit: 4 Evaluation: Practical (P)	LIBRARY HOUSE KEEPING OPERATIONS

COURSE OBJECTIVE

This paper prepares students for their future role as library professionals through theoretical and hands-on activities in the classroom and the RGU Central Library and apart from experience with various categories of libraries in the state as well as country.

COURSE OUTCOMES

CO. No.	Course Outcome	SDG Goals	Bloom's Taxonomy Level
CO1	Define the principles and application of library classification systems.	SDG 4: Quality Education	BT Level I
CO2	Illustrate skills in proper shelving techniques, arranging materials according to classification systems and maintaining a well-organized library layout.	SDG 4: Quality Education SDG 11: Sustainable Cities	BT Level II
CO3	Apply proficiency in using ILMS for automating library housekeeping tasks.	SDG 4: Quality Education SDG 9: Industry & Innovation	BT Level III
CO4	Classify the records, reports, and documentation related to housekeeping operations, including inventory logs, maintenance schedules, and incident reports.	SDG 4: Quality Education SDG 16: Peace, Justice & Strong Institutions	BT Level IV

DETAILED SYLLABUS

Module	Topics and Course Contents	SDG Goals	Periods
Unit 1	Overview of RGU Central Library Operations: Acquisition; Technical; Circulation; Periodical; Reference; Documentation and Maintenance	SDG 4: Quality Education SDG 9: Industry & Innovation	60
Unit 2	E-Resources: A Report on observation of Electronic Resources including Green Open Access, Gold Open Access, Hybrid Resources and Mandate Open Access – submitted for evaluation	SDG 4: Quality Education SDG 10: Reduced Inequalities	
Unit 3	Job Diary: A Report on observation of RGU Central Library operations – submitted for evaluation with viva voce	SDG 4: Quality Education SDG 8: Decent Work & Economic Growth	
Unit 4	Study Tour Diary: A Report on operations of Libraries visited during the study tour – submitted for evaluation with viva voce	SDG 4: Quality Education SDG 17: Partnerships for the Goals	
	TOTAL PERIODS		**60**

TEXTBOOKS

Beardwell, I. & Holden, L. (1996). Human resource management. UK: Longman.
Bryson, J. (1999). Effective library and information management. Bombay: Jaico.

REFERENCE BOOKS

Narayan, G.J. (1991). Principles and practice of management. New Delhi: Prentice Hall of India.
Scammell, Alison. (Ed.). (2001). Handbook of information management. 8th ed. London: Aslib-IMI.

SEMESTER IV

Fourth Semester Courses

Master of Library and Information Science (MLISC)

LIB224C401 L-T-P-C: 0-0-8-4 | Credit: 4 Evaluation: Practical (P)	E-LEARNING AND CONTENT MANAGEMENT SYSTEM

COURSE OBJECTIVE

To train the students on Learning Management and Content Management System using Open-Source Software.

COURSE OUTCOMES

CO. No.	Course Outcome	SDG Goals	Bloom's Taxonomy Level
CO1	Define the concept of e-learning and its various forms (synchronous, asynchronous, blended learning).	SDG 4: Quality Education	BT Level I
CO2	Explain the fundamental concepts of Content Management Systems (CMS).	SDG 4: Quality Education SDG 9: Industry & Innovation	BT Level II
CO3	Apply instructional design principles to create engaging and effective e-learning content.	SDG 4: Quality Education SDG 9: Industry & Innovation	BT Level III
CO4	Analysing learning analytics, and evaluating the effectiveness of e-learning content.	SDG 4: Quality Education	BT Level IV

DETAILED SYLLABUS

Module	Topics and Course Contents	SDG Goals	Periods
Unit 1	E-Learning, Evolution – Generations of distance education; Open-Source CMS: MediaWiki, Joomla, Drupal, Zope; Wiki Hosting, Wiki Technologies in Libraries for Subject-Based Encyclopaedias, Subject directory/portal	SDG 4: Quality Education SDG 9: Industry & Innovation	10
Unit 2	Open-Source LMS: Moodle, A Tutor (Virtual Classroom); Courseware Management; Massive Open Online Courses (MOOCs): Background, Concept; E-content development process	SDG 4: Quality Education	10
Unit 3	Practice on CMS (Joomla/Drupal/WordPress): Installation, Customization, Configuration, Content Publishing, RSS Feed Integration, Designing	SDG 4: Quality Education SDG 9: Industry & Innovation	20
Unit 4	Practice on LMS (Moodle/A-Tutor): Installation, Customization, Configuration, Live Classroom, Designing, Video Lecture Room, Plugin Integration	SDG 4: Quality Education SDG 9: Industry & Innovation	20
	TOTAL PERIODS		**60**

TEXTBOOKS

Boiko, Bob. (2005). Content Management Bible. 2nd Ed. Wiley Publishing.
Buchner, Alex. (2011). Moodle 2 Administration. Packt Publishing.

REFERENCE BOOKS

Burge, Stephen. (2011). Joomla! Explained. Pearson Education.
Dvorak, Radana. (2011). Moodle for Dummies. Wiley Publishing.

LIB224C402 L-T-P-C: 3-1-0-4 | Credit: 4 Evaluation: Theory (T)	DIGITAL RESOURCES MANAGEMENT

COURSE OBJECTIVE

To provide nature, features, scopes, and limitations of digital information resources and also know the use of text retrieval and support tools in organizing digital resources.

COURSE OUTCOMES

CO. No.	Course Outcome	SDG Goals	Bloom's Taxonomy Level
CO1	Define digital resources and their types (e.g., text, images, videos, multimedia).	SDG 4: Quality Education SDG 9: Industry & Innovation	BT Level I
CO2	Explain the importance of effective management of digital resources in education, libraries, etc.	SDG 9: Industry & Innovation	BT Level II
CO3	Apply ethical practices and guidelines for respecting intellectual property rights in digital resource management.	SDG 16: Peace, Justice & Strong Institutions	BT Level III
CO4	Analyse text retrieval engines that support multimedia and cross-media content.	SDG 9: Industry & Innovation	BT Level IV

DETAILED SYLLABUS

Module	Topics and Course Contents	SDG Goals	Periods
Unit 1	Digital information resources: definition, scope, features, advantages; Socio-legal aspects: copyright, IPR issues, licensing	SDG 9: Industry & Innovation SDG 16: Peace, Justice & Strong Institutions	15
Unit 2	Text retrieval engines (Apache-Solr, Zebra); OMEKA (DMS); OJS (Open Journal System); CORAL (ERM system): Features, Plugins, Architecture	SDG 9: Industry & Innovation	15
Unit 3	Web 2.0: components, tools, technologies; Library 2.0 – RSS, Wikimedia, Blog, Social networking, Social bookmarking, Carousel; Information mashup; Trends	SDG 4: Quality Education SDG 9: Industry & Innovation	15
Unit 4	Metadata: types, models, best practices; Dublin Core (Simple and Qualified); Domain-specific: Learning objects, ETD; RDF, XML and Metadata schemas	SDG 4: Quality Education SDG 9: Industry & Innovation	15
	TOTAL PERIODS		**60**

TEXTBOOKS

Akerkar, R. (2009). Foundations of the Semantic Web. Oxford: Alpha Science International.

Arthur, M. H. (2006). Expanding a digital content management system. Amsterdam: Elsevier Focal Press.

REFERENCE BOOKS

Berry, M. W. & Browne, M. (2005). Understanding search engines. Philadelphia: SIAM.

Croft, W. B. et al. (2010). Search engines: Information retrieval in practice. Boston: Addison-Wesley.

LIB224C413 L-T-P-C: 0-0-8-4 | Credit: 4 Evaluation: Practical (P)	DIGITAL RESOURCES MANAGEMENT (PRACTICAL)

COURSE OBJECTIVE

To explore the use of Digital Resource Management Systems and Semantic web technologies in library systems with hands-on practice.

COURSE OUTCOMES

CO. No.	Course Outcome	SDG Goals	Bloom's Taxonomy Level
CO1	Define role of metadata in organizing and describing information resources.	SDG 4: Quality Education	BT Level I
CO2	Summarize metadata standards and schemas such as Dublin Core, MARC, and MODS.	SDG 4: Quality Education SDG 9: Industry & Innovation	BT Level II
CO3	Apply metadata standards and protocols in real-world scenarios to enhance resource discoverability and accessibility.	SDG 9: Industry & Innovation SDG 10: Reduced Inequalities	BT Level III
CO4	Analyse effective search strategies within DRM systems for quick and precise retrieval of resources.	SDG 9: Industry & Innovation	BT Level IV

DETAILED SYLLABUS

Module	Topics and Course Contents	SDG Goals	Periods
Unit 1	Hands-on Practice on OJS (Open Journal System): Features, Plugins, Metadata, Architecture and Modules	SDG 4: Quality Education SDG 9: Industry & Innovation	20
Unit 2	Hands-on Practice on OMEKA: Features, Plugins, Image Metadata, Video Metadata, Architecture	SDG 9: Industry & Innovation	20
Unit 3	Hands-on Practice on CORAL (Electronic Resources Management): Features, Plugins, Architecture and Modules	SDG 9: Industry & Innovation	20
Unit 4	Metadata management; Collection building; Import/Export of Metadata; Backup and Restore; Indexing with search engine and database (Google Scholar/Academic/BASE)	SDG 9: Industry & Innovation SDG 10: Reduced Inequalities	10
	TOTAL PERIODS		**70**

TEXTBOOKS

Akerkar, R. (2009). Foundations of the Semantic Web. Oxford: Alpha Science International.

REFERENCE BOOKS

Berry, M. W. & Browne, M. (2005). Understanding search engines. SIAM.
Croft, W. B. et al. (2010). Search engines: Information retrieval in practice. Addison-Wesley.

LIB224C404 L-T-P-C: 3-1-0-4 | Credit: 4 Evaluation: Theory (T)	DATABASE MANAGEMENT SYSTEM

COURSE OBJECTIVE

This paper is intended to develop a comprehensive understanding of the nature, technological environments, models, and applications of database management system.

COURSE OUTCOMES

CO. No.	Course Outcome	SDG Goals	Bloom's Taxonomy Level
CO1	Define concepts such as data models, schema design, normalization, indexing, and querying languages.	SDG 4: Quality Education SDG 9: Industry & Innovation	BT Level I
CO2	Explain the basic concepts of the relational data model.	SDG 4: Quality Education	BT Level I
CO3	Develop the DBMS software, understanding database architectures, security, and transactions.	SDG 9: Industry & Innovation	BT Level III
CO4	Discover knowledge of database administration: backup, recovery, user management, access control, performance tuning.	SDG 9: Industry & Innovation SDG 16: Peace, Justice & Strong Institutions	BT Level IV

DETAILED SYLLABUS

Module	Topics and Course Contents	SDG Goals	Periods
Unit 1	Introduction, Characteristics; Data Models, Schemas and Instances; Database Architectures: Three Schema, Client/Server; Data Independence; Languages, Interfaces, Environment; Classification of DBMS; Phases of Database Design	SDG 4: Quality Education SDG 9: Industry & Innovation	15
Unit 2	ER Model: Entity Types, Entity Sets, Attributes and Keys; Relationships: Types, Sets, Roles and Constraints; EER Model: Subclasses, Superclasses, Inheritance; Specialization and Generalization	SDG 9: Industry & Innovation	15
Unit 3	Data Abstraction, Knowledge Representation, Ontology; Relational Data Model: Concepts, Constraints, Schemas; Update Operations, Transactions; Functional Dependencies; Normalization; Relational Database Design	SDG 9: Industry & Innovation	15
Unit 4	Object-Oriented Databases: Object Identity, Structure, Type Constructors; Encapsulation, Methods, Persistence; Class Hierarchies, Inheritance; Object DB Standards; Object-Relational and Extended DB Systems; Emerging Technologies	SDG 9: Industry & Innovation	15
	TOTAL PERIODS		**60**

TEXTBOOKS

Elmasri, Ramez & Navathe, Shamkant B. (2011). Fundamentals of Database Systems. 6th Ed. Addison-Wesley.

REFERENCE BOOKS

Hentzen, Whil. (2007). MySQL Client-Server Applications with Visual FoxPro. Hentzenwerke.
Welling, Luke & Laura Thomson. (2003). PHP and MySQL Web Development. 2nd Ed. Sams.

LIB224C405 L-T-P-C: 3-1-0-4 | Credit: 4 Evaluation: Theory (T)	MEDIA INFORMATION LITERACY AND COPYRIGHT

COURSE OBJECTIVE

The objective of the course is to acquire knowledge on information literacy in the context of different media and to acquire knowledge of copyright.

COURSE OUTCOMES

CO. No.	Course Outcome	SDG Goals	Bloom's Taxonomy Level
CO1	Define on media and information literacy model.	SDG 4: Quality Education SDG 16: Peace, Justice & Strong Institutions	BT Level I
CO2	Explain information literacy for their academic work.	SDG 4: Quality Education	BT Level II
CO3	Develop the concept of Intellectual Property Right and copyright.	SDG 16: Peace, Justice & Strong Institutions	BT Level III
CO4	Compare the guidelines and Standards: UNESCO, IFLA and ALA for Information Literacy.	SDG 4: Quality Education SDG 17: Partnerships for the Goals	BT Level IV

DETAILED SYLLABUS

Module	Topics and Course Contents	SDG Goals	Periods
Unit 1	Information Literacy: Definition, Models and Standards; Strategic Plan; Information Literacy and Lifelong Learning; Information Society and Information Literacy	SDG 4: Quality Education SDG 16: Peace, Justice & Strong Institutions	15
Unit 2	Media Literacy, Computer Literacy, Information Literacy and Media Literacy; Media Literacy and Bridging the Digital Divide; Media Literacy: Impact on Academic, Public, and Special Library	SDG 4: Quality Education SDG 10: Reduced Inequalities	15
Unit 3	Media and Information Literacy: Major Initiatives in USA, UK and Australia; Policies, Guidelines and Standards: UNESCO, IFLA and ALA; MIL: Skills and Competencies; MIL: Best Practices	SDG 4: Quality Education SDG 17: Partnerships for the Goals	15
Unit 4	Intellectual Property Rights IPR: Concept, Genesis, Development and Categories; Digital Rights Management; IPR Acts and its Application in Electronic Environment; Violation and Infringement of IPR; Copyright and Patents; Patent Laws in India and Abroad	SDG 16: Peace, Justice & Strong Institutions SDG 9: Industry & Innovation	15
	TOTAL PERIODS		**60**

TEXTBOOKS

Ainley, P. (1980). Basics of community information. London: AAL.

American Library Association. (1966). Minimum standards for public library systems. Chicago: ALA.

REFERENCE BOOKS

Bunch, A. (1982). Community information services. London: Clive Bingley.

Durrance, J. C. & Fisher, K. E. (2002). Online community information. Chicago: ALA.